

Date: 02 July 2026	Team ID: SB-7429
Project Name: Credit Card Approval Prediction using Machine Learning	Maximum Marks: 1 Mark

SCALABILITY & FUTURE PLAN TEMPLATE

Current System Limitations

S.No	Limitation Identified	Operational Impact	Priority to Address (High/Medium/Low)
1	The dataset contains ~31% missing values in Occupation Type.	Reduces predictive signal strength of the occupation categorical parameter.	Medium
2	Model does not explain why an ML decision was made (black box).	Risk officers cannot explain borderline rejections to clients or auditors.	High
3	Stateless Flask server does not persist historical application logs.	Underwriters cannot audit past decisions or track daily volumes.	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load Capacity	Single-instance local Flask server handling sequential requests.	Deploy inside containerized Docker pods managed by Kubernetes on AWS.
2	Data Storage	Local cache using raw static CSV files for ETL training.	Migrate to a relational cloud database (PostgreSQL) with indexing on ID.
3	Inference Speed	Synchronous CPU-based Random Forest execution (~42ms).	Implement Redis session caching for repeat applicants to achieve <5ms speeds.
4	Security & Access	HTTP local connection with form-encoded inputs.	Enforce HTTPS connection, API Gateway access keys, and JWT authorization.

Future Enhancements Roadmap

Phase / Version	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2 (v2.1)	Explainable AI (SHAP / LIME explanation graphs integration)	Q3 2026	Generates visual feature contribution charts for risk officer auditing.
Phase 3 (v2.2)	Alternative Data Streams integration (utility bills, mobile data)	Q1 2027	Enables credit scoring of thin-file applicants lacking depth history.
Phase 4 (v2.3)	Continuous model training loops & automated drift monitors	Q3 2027	Keeps model calibrated against shifting economic states and risk trends.